

AISHWARYA VIJAYAN

774-357-9709 • aishwaryav88@gmail.com•

EDUCATION

University of Illinois at Chicago May 2025
Doctor of Philosophy in Computer Science, 3.2 GPA

Anna University India May 2011
Master of Science in Computer Science, 3.1 GPA

Kerala University India May 2009
Bachelor of Technology in Computer Science Engineering, 3.0 GPA]

SKILLS

- Programming: C, C++, Python, Java, MATLAB
- Machine Learning: Feature engineering, modeling, evaluation, data mining
- Tools: NLP frameworks, scientific computing, graph/network analysis
- Other: Strong communication, mentoring, curriculum development

TEACHING EXPERIENCE

Roosevelt University Chicago January 2025–Present
Adjunct Faculty

- Teach graduate level and undergraduate level courses in Data Mining and Operating Systems.
- Develop instructional materials and support graduate-level learning.

University of Illinois Chicago
Teaching Assistant Fall 2015, Spring & Fall 2016-2021

- Database Systems and Advanced Database Management Systems under Prof. Ouri Wolfson.
- Programming in C/C++ under Prof. Joe Hummel
- Database systems under Prof. Aravinda Prasad Sistla

SCT College of Engineering India May 2012–June 2013
Assistant Professor (Ad Hoc)

- Taught courses in C Programming and Database Design.

MES College of Engineering India July 2011– May 2012
Assistant Professor

- Taught undergraduate/graduate courses in Database Design and Data Structures.
- Mentored students in research projects.

RESEARCH EXPERIENCE

University of Illinois Chicago

Graduate Research Assistant

August 2022 – January 2023

- Conducted research on human brain data with a focus on understanding mechanisms related to consciousness.
- Performed structured data extraction from unstructured textual sources to support analytical workflows.
- Studied and implemented various prompt engineering techniques for optimizing language model performance.
- Analyzed and compared the effectiveness of prompt engineering methods across different large language models.

Graduate Research Assistant

Summer 2016, 2017, 2018

- Conducted data-driven research on human brain networks using traffic-assignment models such as System Optimum and User Equilibrium
- Investigated coordination and congestion in brain networks under unconscious states using structural and functional connectomes.
- Developed Markov Decision Process (MDP)–based models to analyze dynamic signal routing in the brain

SSN College of Engineering India

January 2011-March 2011

Research Assistant

- Designed IP-based WSN architecture using 6LoWPAN to support remote sensor access via HTTP without translation layers.

PUBLICATIONS

- V. Aishwarya and V.S.F. Enigo, “IP based wireless sensor networks with web interface,” ICRTIT 2011.
- O. Wolfson, P. Szczurek, A. Vijayan et al., “A Traffic Analysis Perspective on Communication in the Brain,” MDM 2017.
- A. Vijayan, “Deciphering Neural Codes: A Resource Search Network Perspective on Brain Connectivity,” ACAI 2023.
- A. Vijayan, “A Prompt Engineering Approach for Structured Data Extraction from Unstructured Text,” ACAI 2023.

REFERENCES

- Prof. Ouri Wolfson, UIC, Computer Science owolfson@gmail.com
- Prof. A. Prasad Sistla, UIC, Computer Science sistla@uic.edu
- Prof. Anastasios Sidiropoulos, UIC Computer Science sidiropo@gmail.com
- Prof. Olusola Ajilore, Psychiatry, University of Illinois Chicago oajilore@uic.edu
- Prof. Alex Leow, Psychiatry & Bioengineering UIC alexfeuillet@uic.edu
- Dr. Jerry Schnepf, Roosevelt University, jschnepf@roosevelt.edu